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HRB HRB 6S 4000mAh 60C XT90 LiPo Battery

HRB 6S Lipo Battery User Manual

Model: 4000mAh 60C 22.2V XT90

1. Product Overview

The HRB 6S Lipo Battery is a high-performance lithium polymer battery designed for various RC applications, including quadcopters, helicopters, cars, trucks, and boats. This manual provides essential information for safe and effective use, ensuring optimal performance and longevity of your battery.



Figure 1: HRB 6S 4000mAh 60C LiPo Battery (2-pack).

[redacted]

Video 1: An overview of HRB Lipo Battery manufacturing and application in RC models.

2. Safety Guidelines

Lithium Polymer (LiPo) batteries require careful handling to prevent damage, fire, or injury. Adhere to the following safety precautions:

- **Charging:** Always use a LiPo compatible charger with a balance function. Never charge unattended. Do not charge batteries that are swollen, damaged, or have been physically impacted. The safest charge rate for LiPo batteries is 0.5C-1C (2A-4A for this 4000mAh battery).
- **Discharging:** Do not over-discharge the battery. Stop use immediately if the voltage drops below 3.7V per cell. Over-discharging can permanently damage the battery.
- **Storage:** Store LiPo batteries at room temperature (15-25°C or 59-77°F) in a fireproof container or bag, away from flammable materials. For long-term storage, charge or discharge the battery to storage voltage (typically 3.8V-3.85V per cell).
- **Physical Damage:** Inspect the battery surface for any damage before each use. Do not use a battery that is punctured, cut, or shows signs of swelling.
- **Temperature:** Avoid exposing batteries to extreme temperatures (above 60°C/140°F or below 0°C/32°F).
- **Disposal:** Dispose of damaged or old LiPo batteries responsibly according to local regulations. Never dispose of them in regular trash.

3. Product Specifications

Feature	Specification
Chemistry	Li-Polymer
Cells Number	6S
Voltage	22.2V
Capacity	4000mAh
Discharge Rate	60C
Burst Discharge Rate	120C
Connector Type	XT90
Style	Soft Case
Dimensions (L*W*H)	135*42*49mm (5.31*1.65*1.93in)
Weight	590g (1.3lb / 20.81oz)

Key Features:

- A+ Grade Cell for enhanced safety and performance.
- Longer Running Time and Lower Resistance.
- Extended Cycle Life: Up to 350+ times under correct use and maintenance.
- Lighter Weight design.
- High Energy Density: Approximately 320Wh/kg.

Figure 2: Advanced Battery Performance features.

Figure 3: Physical dimensions and connector details of the battery.

4. Setup and Installation

Before installing the battery, ensure your RC model is compatible with 6S (22.2V) LiPo batteries and has an XT90 connector. Verify that the dimensions of the battery fit securely within your model's battery compartment.

1. **Prepare Model:** Turn off the power of your RC car, quadcopter, or other model.
2. **Insert Battery:** Carefully place the HRB LiPo battery into the battery compartment of your model. Ensure it is seated firmly and does not shift during operation.
3. **Secure Battery:** Fasten the battery using straps or other securing mechanisms provided with your model to prevent movement.
4. **Connect Plugs:** Connect the XT90 power plug from the battery to the corresponding connector on your RC model. Ensure the connection is secure and correct. If your model uses a different connector, use a compatible adapter or replace the connector with an XT90.
5. **Balance Lead:** Connect the 7-pin balance port to your balance charger or flight controller if applicable.

Figure 4: Easy steps to connect the HRB LiPo battery.

5. Operating Instructions

5.1. Charging the Battery

Always charge your HRB LiPo battery using a dedicated LiPo balance charger. This ensures that all cells are charged equally, which is crucial for battery health and safety.

- Connect the main XT90 plug to the charger's output.
- Connect the balance lead to the charger's balance port.
- Set the charger to the correct battery type (LiPo), cell count (6S), and capacity (4000mAh).
- Set the charging current. For optimal battery life, a charge rate of 0.5C to 1C is recommended. For this 4000mAh battery, 0.5C is 2A and 1C is 4A.
- Start the charging process and monitor the battery. Discontinue charging immediately if you notice any unusual heat, swelling, or smoke.



Figure 5: Proper charging connection and recommended charge rate.

5.2. Discharging the Battery

The HRB 6S 4000mAh 60C LiPo battery is designed for high power output, suitable for demanding RC applications. Its 60C continuous discharge rate and 120C burst rate provide ample power for your models.

- Ensure your RC model's ESC (Electronic Speed Controller) is configured for LiPo batteries and has a low voltage cutoff (LVC) set appropriately for 6S LiPo (e.g., 3.4V-3.7V per cell).
- Monitor battery voltage during use. Stop operation immediately if you notice a significant drop in power or if the LVC is triggered.
- Avoid over-discharging the battery below 3.7V per cell, as this can cause irreversible damage.



Figure 6: High power output for RC applications.

5.3. Storage

For short-term storage (a few days), you can leave the battery fully charged. For long-term storage (more than a week), it is crucial to store LiPo batteries at their storage voltage (3.8V-3.85V per cell). Most modern LiPo chargers have a "storage" mode for this purpose.

- Store batteries in a cool, dry place, away from direct sunlight and heat sources.
- Use a fireproof LiPo bag or container for added safety during storage.
- Do not store fully charged or fully discharged batteries for extended periods.

6. Maintenance

Proper maintenance extends the life and performance of your HRB LiPo battery.

- **Regular Inspection:** Before and after each use, visually inspect the battery for any signs of damage, swelling, or punctures.

Check connectors for wear or corrosion.

- **Temperature Management:** Allow the battery to cool down to ambient temperature before recharging after use. Avoid operating the battery if it becomes excessively hot.
- **Balance Charging:** Always balance charge your battery to ensure cell voltages remain consistent. This prevents individual cells from becoming overcharged or undercharged, which can lead to premature degradation.
- **Cycle Life:** The HRB LiPo battery is designed for a long cycle life, up to 350+ cycles, provided it is used and maintained correctly.



Figure 7: Long cycle life and high energy density.

7. Troubleshooting

If you encounter issues with your HRB LiPo battery, consider the following common problems and solutions:

Problem	Possible Cause	Solution
Battery not charging	Incorrect charger settings, damaged connectors, faulty charger, deeply discharged battery.	Verify charger settings (LiPo, 6S, 4000mAh). Check all connections. Try a different charger. If deeply discharged, some chargers have a "LiPo revive" mode, but proceed with caution.
Battery swelling/puffing	Overcharging, over-discharging, physical damage, excessive heat, age.	**Immediately discontinue use.** This indicates internal damage and a fire risk. Safely discharge the battery to 0V and dispose of it according to local regulations. Do not attempt to use or recharge.
Short run time	Battery degradation, improper charging, high current draw from model, cold temperature.	Ensure proper charging and balance. Check your model's power system for excessive current draw. Operate in suitable temperatures. Consider battery replacement if degradation is significant.
Battery gets hot during use	Excessive current draw, internal resistance, over-discharging.	Reduce current draw (e.g., use a smaller propeller, less aggressive driving). Ensure battery is adequately rated for your application (60C continuous). Allow battery to cool before next use.

8. Warranty and Support

HRB is committed to providing high-quality products. If you experience any problems with your HRB 6S Lipo Battery, please contact HRB customer support immediately. You can typically find contact information through your purchase platform (e.g., "Your Orders" section on Amazon) or by sending an email to the after-sales mailbox provided with your product documentation. HRB aims to reply to inquiries within 24 hours to assist with any product-related concerns.

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